

# Shivanshu Mishra

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## EDUCATION

SUNY Buffalo | MS in Artificial Intelligence | New York

June 2024 – Dec 2025

University of Mumbai | BE in AI and ML | India

Aug 2020 – May 2024

## WORK EXPERIENCE

*AI Engineer Intern – Samtek – Virginia*

*June 2025 – August 2025*

- Used Python, AWS Bedrock, and NLP to develop a robust scalable, AI-powered cloud configuration system that automatically converts cloud setup files, reducing manual effort by 30 hours per week.
- Designed a Stream-lit based app using FastAPI and CI/CD pipeline for file handling, conversion and validation using LLMs across different cloud platforms and used Power BI dashboards for real-time process monitoring.
- Created a dual-interface platform accessible to technical and non-technical users alike, integrating web and CLI access to simplify workflows and boost cross-team adoption.
- Implemented a caching framework using Python OOP and Machine Learning libraries (NumPy, Pandas, scikit-learn), cutting API calls by 80% and enabling faster cache tracking.

*Technical Lead Intern – Alternative Clinic – Mumbai*

*June 2023 – September 2023*

- Led development of a predictive data model using Pandas and Scikit-learn, reducing false positives by 15% and making it possible for patients to get treatment earlier.
- Worked in a client-facing role and launched a digital clinic portal using Flask and AWS (S3) for storage with appointment booking and health record access, reducing admin workload.

*AI and ML Intern – VLine Infotech Pvt Ltd – Delhi*

*January 2023 –April 2023*

- Framed and enhanced an ML pipeline using TensorFlow, OpenCV, and Tesseract OCR to extract text from bank statements, achieving a 15% accuracy improvement through iterative testing and fine-tuning.
- Built an ETL pipeline for financial documents by fine-tuning LayoutLM (Hugging Face) and deploying with Kubernetes, improving parsing speed 20% and ensuring reliable key information extraction.

## PROJECTS

*Cheque Image Feature Extraction System ([Link](#))*

- Worked with deep learning models (TrOCR, Donut, and Tesseract) to build OCR systems for handwritten financial checks, using transfer learning, and maintained a Git repository for teamwork.
- Used Computer vision techniques (resolution normalization, ROI cropping) with OpenCV and matplotlib to preprocess 14k+ check images and created a front end and deployed on GCP for user interaction.

*Multi-Modal Movie Genre Classification ([Link](#))*

- Implemented a multi-modal AI system combining NLP (DistilBERT) and computer vision (ConvNeXt-tiny, Torchvision) to classify movies, trained on large datasets and optimized for scalable performance.
- Developed and deployed a cloud-ready full-stack web application with Flask and REST APIs, implementing CI/CD pipelines, scalable data handling, and performance optimizations.

## SKILLS

**Programming & Tools:** Python, SQL, GitHub, AWS, GCP, Power BI

**ML & AI Frameworks:** TensorFlow, PyTorch, Scikit-learn, Hugging Face, XGBoost, OpenCV, MLflow

**Specialized Techniques:** Computer Vision, NLP, CNNs, OCR, Transfer Learning, PCA, t-SNE, GridSearchCV